

Excelerate Connect - Week 4 Final Deliverable

Mobile App Development with Flutter Virtual Internship

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Internship	Mobile App Development with Flutter Virtual Internship
Deliverable	Week 4 - Final application, testing, documentation, and presentation evidence
Application	Excelerate Connect
Repository	github.com/franciskwarteng677/excelerate-connect

[Open the GitHub repository](https://github.com/franciskwarteng677/excelerate-connect)

Final local functional prototype - bundled JSON catalogue, client-side validation, and simulated form submissions only.

Week 4 Objective

Week 4 focused on finalising and presenting the Excelerate Connect Flutter prototype as a coherent, professionally documented project. The work built on the Week 1 proposal and wireframes, the Week 2 responsive static user interface, and the Week 3 dynamic catalogue and functional local forms. The final stage concentrated on refinement, reliability, accessibility, responsive behaviour, manual testing, and submission evidence.

Final Project Overview

Excelerate Connect is a learner-facing mobile application prototype designed to centralise the discovery of sample learning programs and provide clear routes into program details, registration, and feedback. It demonstrates how a learner could browse opportunities through a consistent Flutter interface while retaining accurate limitations: there is no live authentication, backend, database, API, or persistent submission service.

Problem Being Addressed

Learners need a simple way to discover programs, compare details, and interact with application or feedback forms. The prototype also explores how a future platform could organise announcements, learner activity, and administrative workflows. In the current version, these broader capabilities remain planned rather than implemented.

Target Users

- Learners who want to browse sample programs, review eligibility and delivery details, and test registration or feedback flows.
- Future administrators who may later manage programs, announcements, learner participation, and submitted feedback once a backend is introduced.

Final Scope

Area	Final implemented behaviour
Login	Client-side email and password validation; no real authentication or account service.
Home	Responsive landing screen with featured programs loaded through the shared repository.
Program Listing	Four bundled sample programs with search, category filters, result counts, Reset, and empty state.
Program Details	Selected program title, category, descriptions, deadline, duration, delivery format, eligibility, and form actions.
Registration	Validated local form with progress state, simulated submission, success confirmation, and form clearing.
Feedback	Validated local form with rating and comments, simulated submission, success confirmation, and form clearing.
Responsive Design	Layouts manually checked at mobile, tablet, and larger viewport dimensions.

Development Progression

Week 1	Defined the problem, target users, application proposal, user flows, and wireframes.
Week 2	Implemented the responsive Flutter project structure and static UI screens for Login, Home, Program Listing, and Program Details.
Week 3	Added bundled JSON data, typed program parsing, repository-driven loading, search/filtering, selected-program details, validated registration and feedback forms, loading states, and simulated submissions.
Week 4	Audited and refined the application, improved validation and accessibility, corrected responsive edge cases, expanded automated checks, completed manual Chrome testing, and prepared final documentation and presentation evidence.

Data Integration and State Flow

The application uses a bundled local JSON catalogue rather than a live API. The production data path is:

1. assets/data/programs.json
2. Flutter rootBundle asset loading
3. AssetProgramRepository
4. Program.fromJson validation and typed Program objects
5. Home featured programs and Program Listing
6. Selected Program passed to Program Details
7. Registration or Feedback form opened in the selected program context

Loading, success, source-empty, filtered-empty, and error states are handled explicitly. Search and category selections are retained when a learner opens a program and returns to the listing.

Form Behaviour

- Registration validates full name, email format, education level, a minimum-length reason for joining, and acknowledgement of the local-prototype limitation.
- Feedback validates full name, email format, experience rating, and minimum-length comments.
- Invalid submissions show inline errors and a clear notification, while corrected fields revalidate automatically.
- Valid submissions display progress, prevent duplicates, show local-only success confirmation, and clear the form after completion.
- No registration or feedback data is transmitted to a server or saved persistently.

Week 4 Refinements

- Replaced default Flutter web, PWA, and Android labels with accurate Excelerate Connect prototype metadata.
- Removed an unnecessary portrait-only PWA restriction and improved behaviour on wider screens.
- Removed misleading interaction indicators from non-interactive sample event rows.
- Prevented negative Login layout constraints in extremely short browser windows.
- Improved registration and feedback error correction, automatic revalidation, error notifications, scrolling toward validation errors, and accessibility announcements for success panels.
- Improved loading, empty, error, and Retry semantics and prevented whitespace-only searches from incorrectly activating Reset.
- Added duplicate program-ID detection during JSON parsing and expanded responsive automated tests.

Testing and Validation Results

- dart format passed: 24 Dart files checked with no remaining changes required.
- flutter analyze passed with no issues found.
- flutter test passed with all 42 tests successful.
- flutter build web passed and created a local release build.
- Local build smoke checks returned successful responses for the HTML, manifest, JavaScript, and bundled JSON resources.
- Responsive automated checks passed at 360 x 800 with 140% text scaling, 768 x 1024, 1440 x 900, and a 360 x 48 short-window edge case.
- Protected Week 1-3 documents and evidence were preserved unchanged during the Week 4 application audit.

Manual Chrome Verification

The screenshots in this deliverable document the completed manual test pass. Login validation, valid login navigation, Home loading, program browsing, search, Reset, empty state, selected-program details, registration, feedback, and responsive layouts were all exercised successfully.

Current Limitations

- Login remains client-side validation only; there is no real authentication, account creation, password recovery, or session service.
- Program entries are bundled sample JSON data, not live or currently available offerings.
- Registration and feedback submissions are temporary local simulations and are not transmitted or stored.
- No API, backend, database, cloud storage, analytics service, or persistent local storage is connected.
- Events and announcements remain static samples.
- Progress, Profile, Notifications, account management, and administrator workflows remain planned.
- The loading and submission delays are intentional demonstrations of application state.

Recommended Future Development

- Connect the program catalogue to a verified API or administrative content-management backend.
- Add secure authentication, account recovery, session handling, and role-based learner or administrator access.
- Persist registration and feedback data with appropriate validation, privacy controls, and audit trails.
- Implement learner profiles, progress tracking, notifications, saved opportunities, and announcement management.
- Add analytics and monitoring only after consent, privacy, and data-governance requirements are defined.
- Expand accessibility testing with keyboard-only navigation, screen-reader review, and real-device mobile testing.

Screenshot Gallery

The following twelve screenshots provide the final visual evidence for the Week 4 application. They document functionality and responsiveness in numerical order. The screens contain local illustrative program data only.

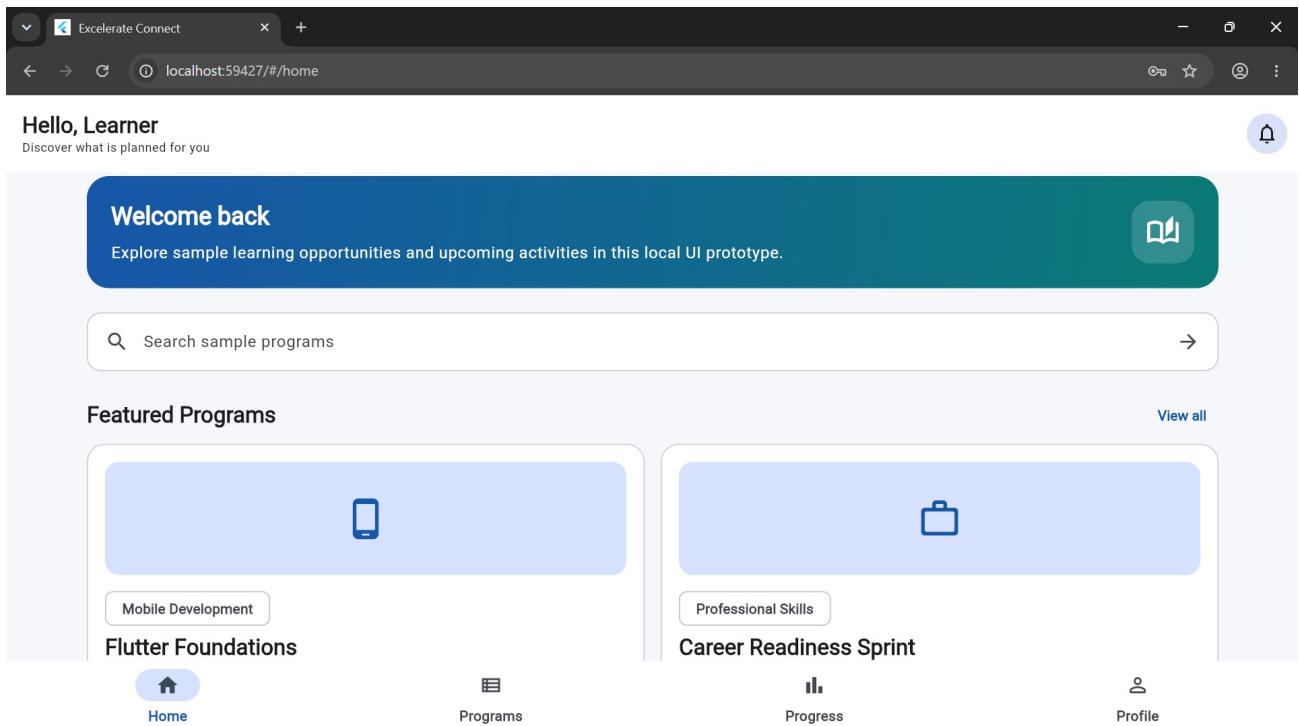
1. Login Validation

The screenshot shows a web browser window with the URL `localhost:59427`. The page displays the 'Excelerate Connect' login interface. At the top, there is a blue icon of a graduation cap. Below it, the text 'Excelerate Connect' is followed by 'Sign in to explore the learner experience.' The form contains two input fields: 'Email address' with a red outline and the placeholder text 'Enter your email address.', and 'Password' with a red outline, a lock icon, a toggle eye icon, and the placeholder text 'Enter your password.'. A 'Forgot password?' link is positioned to the right of the password field. A blue 'Login' button is located below the fields. At the bottom, there is a link for 'New learner? Create account' and a footer note: 'Local UI prototype • No real authentication'.

Caption: Login screen showing required-field validation for email and password.

This screenshot demonstrates that an empty Login submission is blocked and that each required control receives a clear inline error. The footer accurately states that the screen is a local UI prototype with no real authentication.

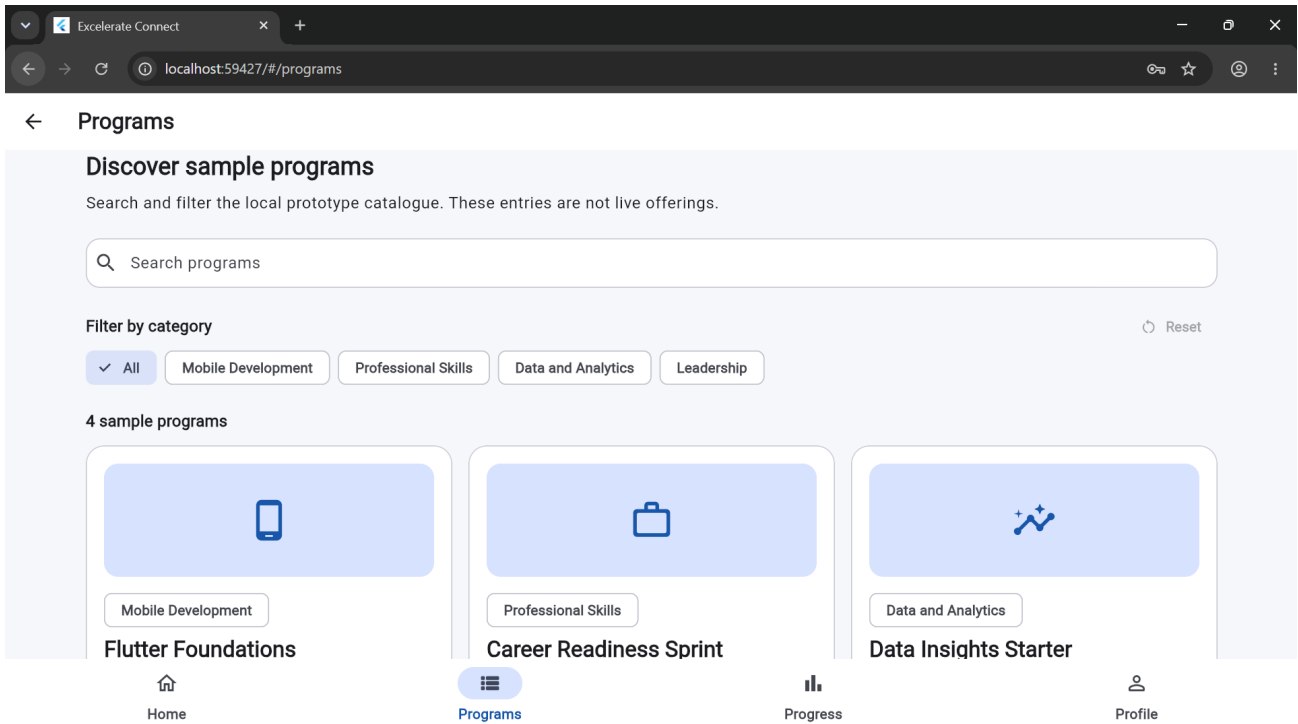
2. Home - Final Responsive View



Caption: Home screen with repository-backed featured programs and responsive navigation.

The Home screen presents the welcome banner, search entry point, featured program cards, notification placeholder, and bottom navigation. The featured records are loaded from the shared bundled JSON data source.

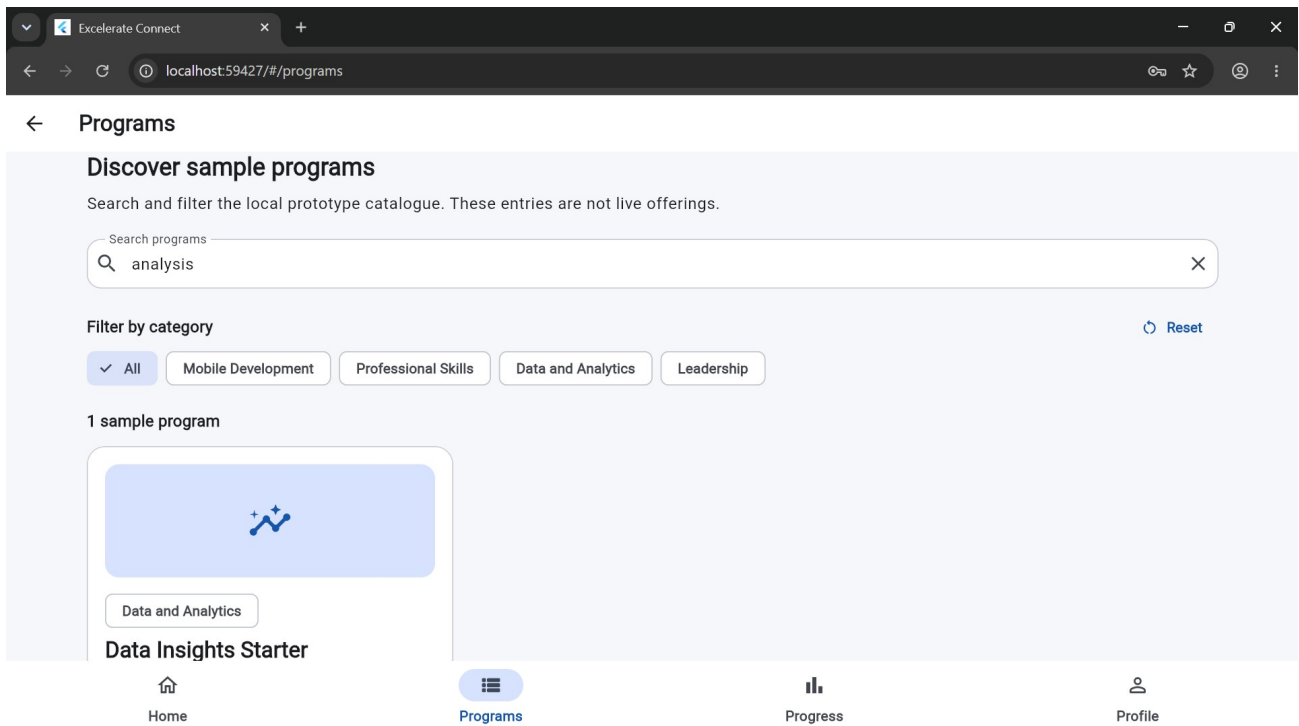
3. Program Listing - Full Catalogue



Caption: Program Listing displaying all four bundled sample programs and category filters.

This screenshot confirms the final catalogue view, filter controls, Reset state, dynamic result count, responsive card layout, and persistent bottom navigation.

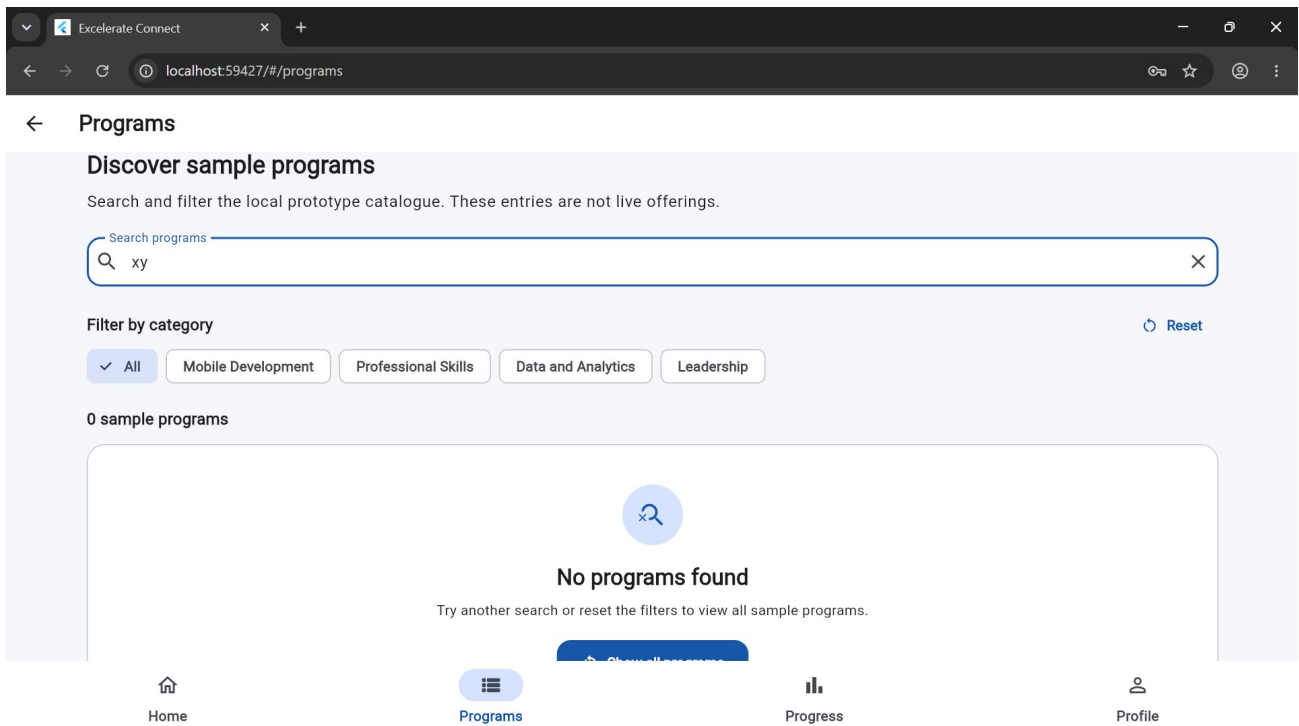
4. Search Result



Caption: Search for "analysis" returning the matching Data Insights Starter program.

The listing updates to one result when the search term matches program content. Reset becomes available because an active search is being applied.

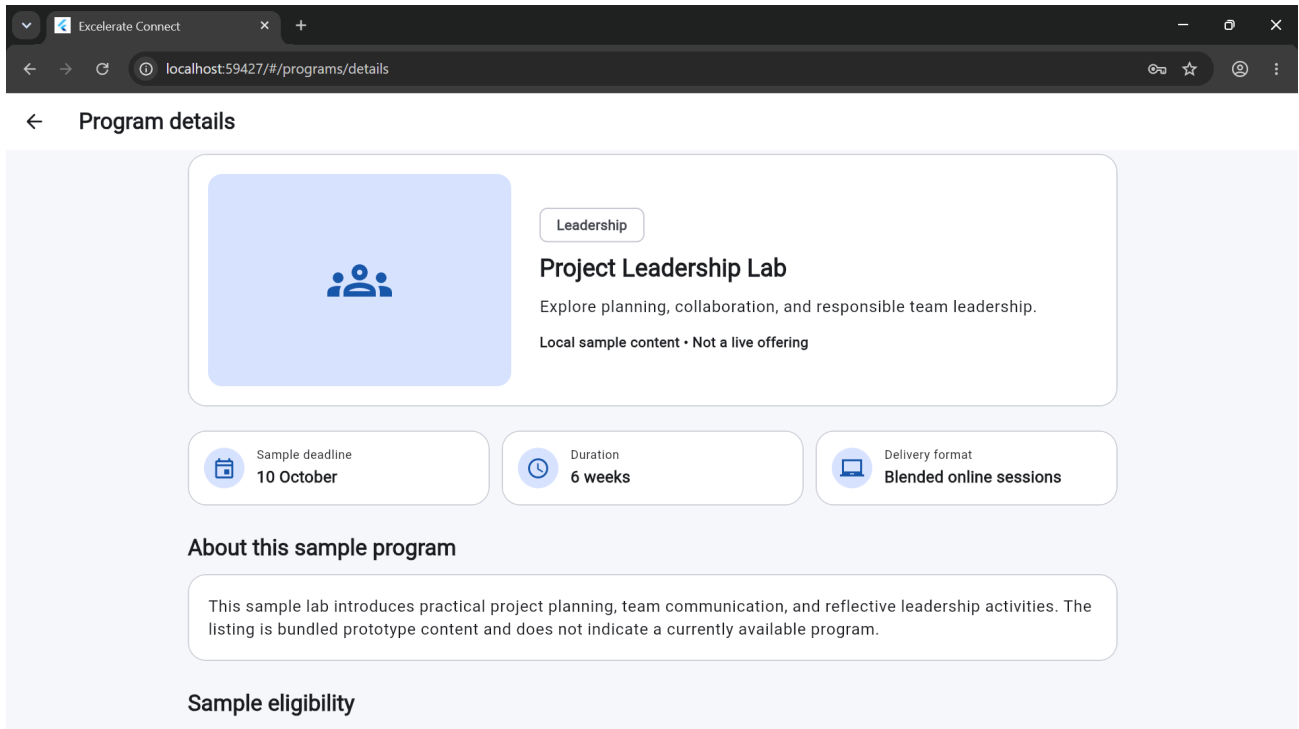
5. Filtered Empty State



Caption: Unmatched search showing a clear zero-result state and recovery action.

This view demonstrates the explicit filtered-empty state. It reports zero programs, explains how to recover, and offers a visible route to restore all sample programs.

6. Program Details - Overview



Caption: Selected Project Leadership Lab program with identity, metadata, and description.

The details screen displays the selected program object rather than a hardcoded default. Category, title, summary, deadline, duration, delivery format, and full description remain consistent with the listing.

7. Program Details - Eligibility and Actions

← Program details

About this sample program

This sample lab introduces practical project planning, team communication, and reflective leadership activities. The listing is bundled prototype content and does not indicate a currently available program.

Sample eligibility

- ✓ Interest in project coordination or team leadership.
- ✓ Availability for collaborative sample exercises.
- ✓ Commitment to respectful and inclusive teamwork.

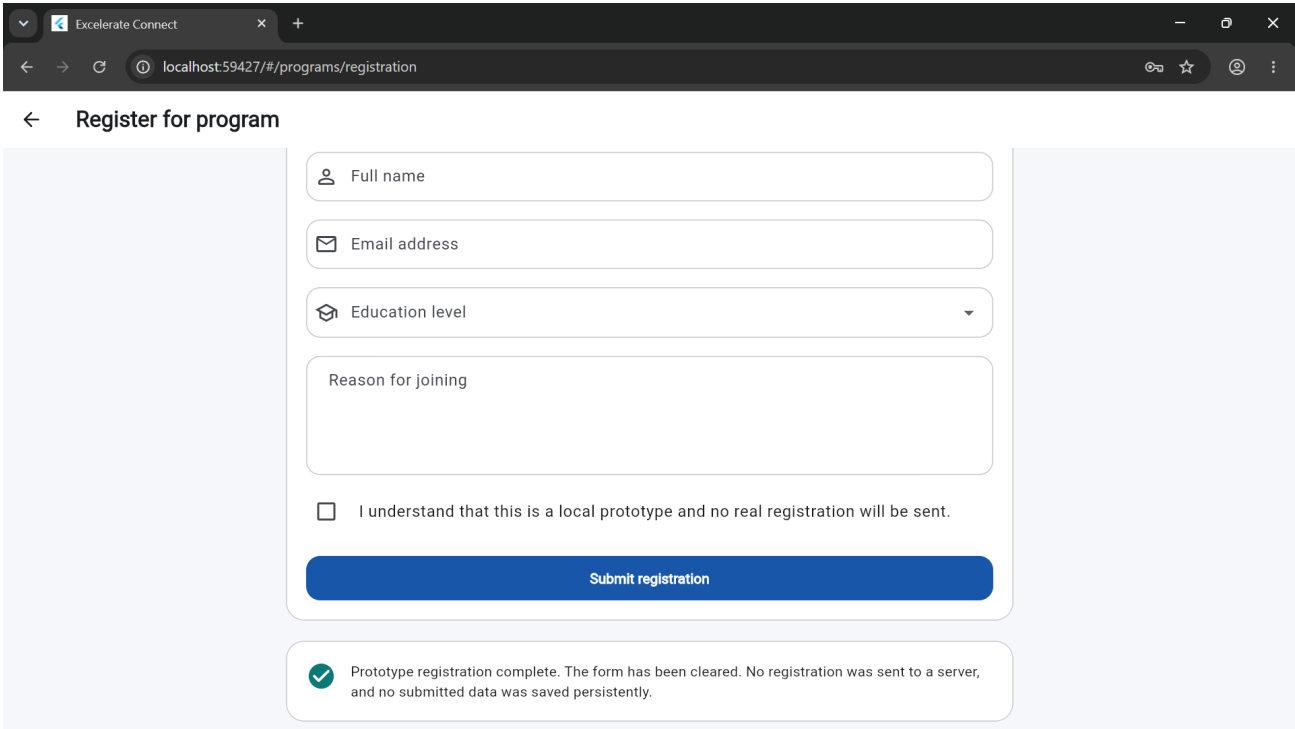
ⓘ Registration and feedback use validated local forms with simulated submissions. Nothing is sent to a server or saved persistently.

[Apply or register](#) [Give feedback](#)

Caption: Eligibility requirements and the Apply or register and Give feedback actions.

The lower details view presents the selected program eligibility list, a truthful local-submission notice, and the two form entry points used during final testing.

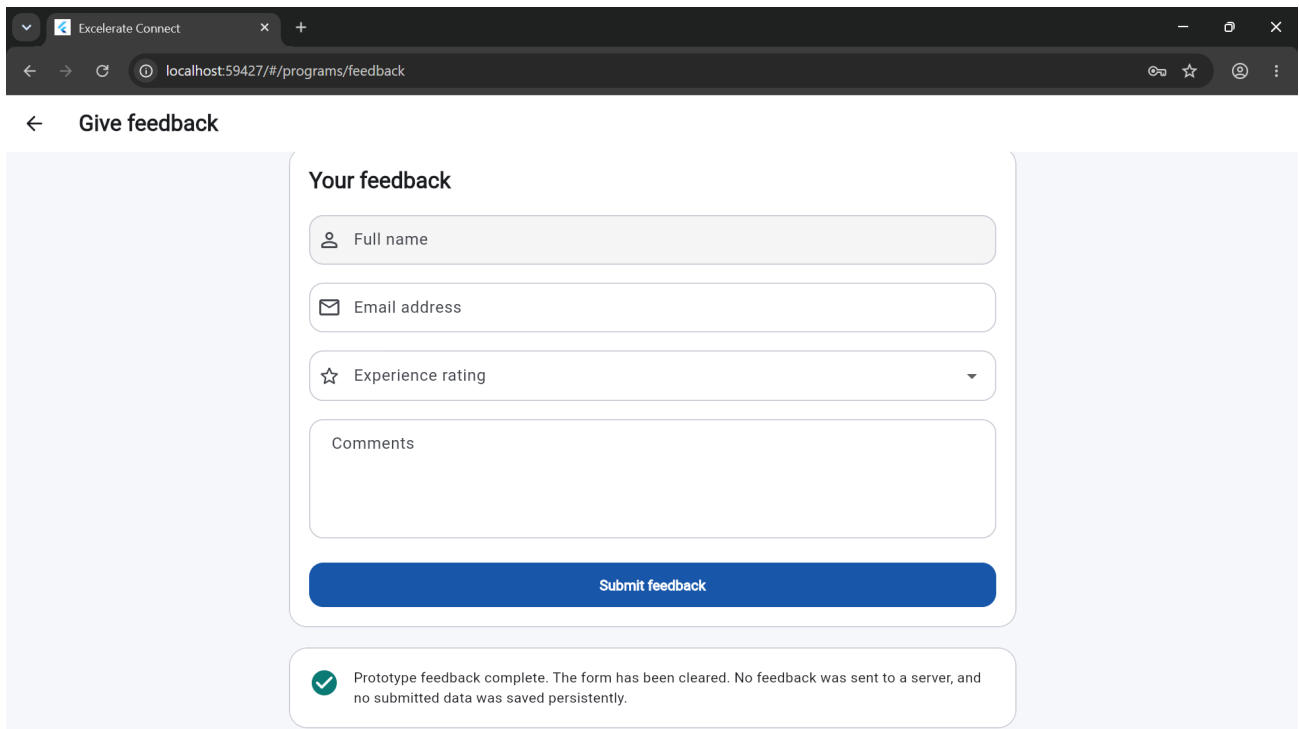
8. Registration Success



Caption: Cleared registration form with persistent local-only success confirmation.

This screenshot records successful simulated registration behaviour. The form was validated, submitted through the temporary local state, cleared afterward, and confirmed without claiming server transmission or persistence.

9. Feedback Success



The screenshot shows a web browser window with the address bar displaying 'localhost:59427/#/programs/feedback'. The page title is 'Give feedback'. The form is titled 'Your feedback' and contains the following fields:

- Full name (text input)
- Email address (text input)
- Experience rating (dropdown menu)
- Comments (text area)

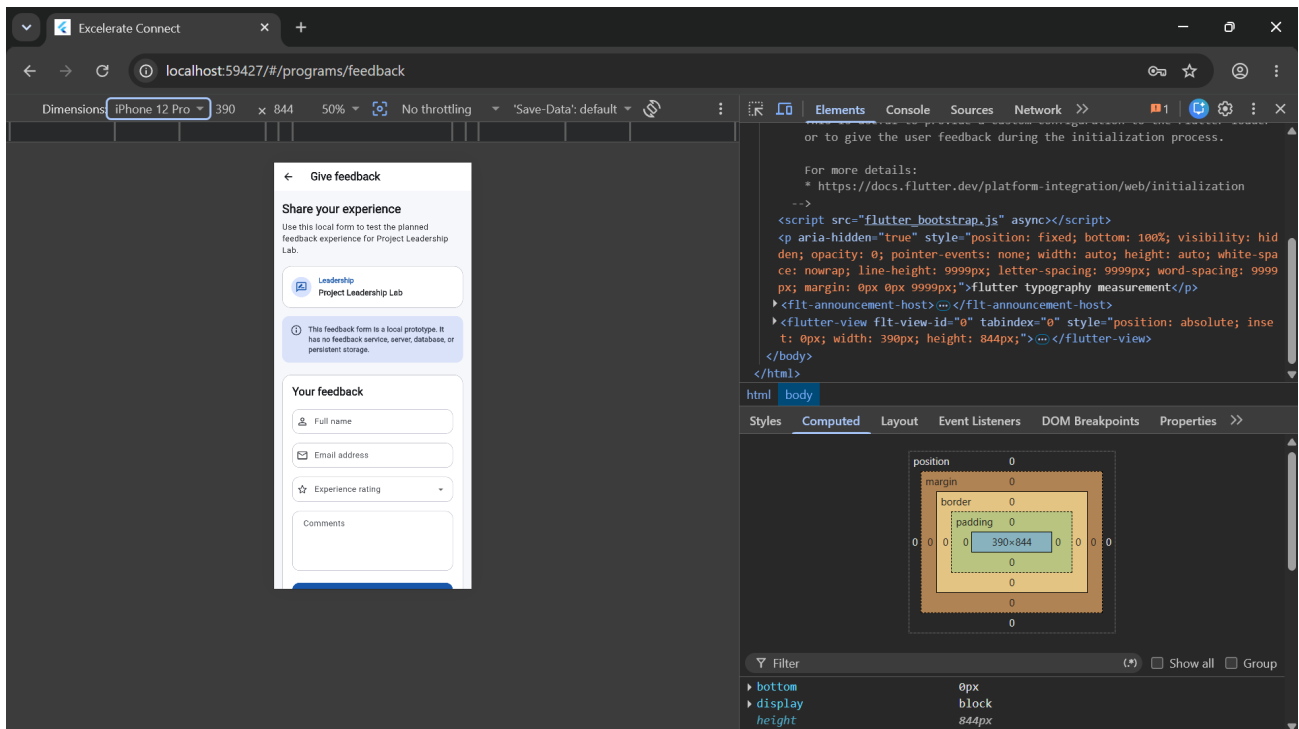
A blue 'Submit feedback' button is located below the form fields. Below the button, a success message is displayed with a green checkmark icon:

Prototype feedback complete. The form has been cleared. No feedback was sent to a server, and no submitted data was saved persistently.

Caption: Cleared feedback form with persistent local-only success confirmation.

The feedback form follows the same reliable pattern: validation, temporary progress, duplicate prevention, success confirmation, and complete clearing of entered values.

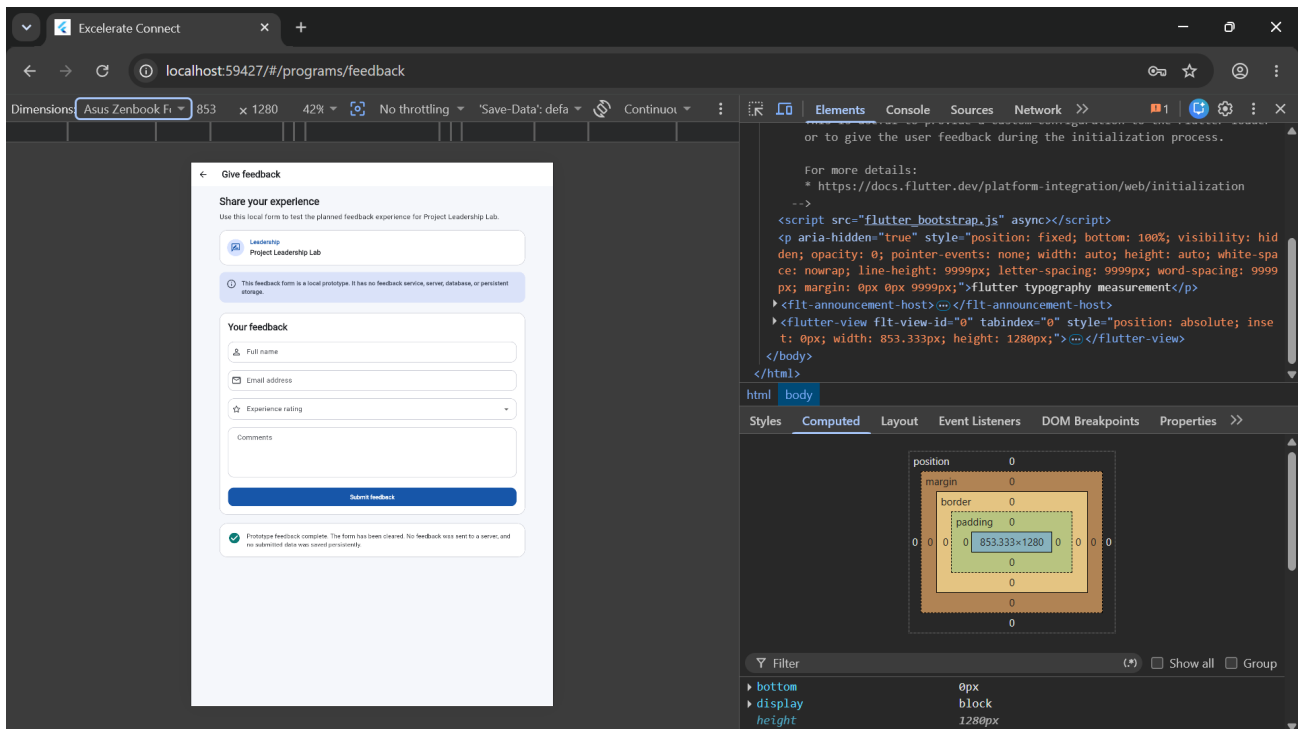
10. Responsive Mobile Check



Caption: Feedback screen tested at an iPhone 12 Pro viewport of approximately 390 x 844.

The narrow mobile layout remains readable and scrollable, with the program context, information panel, form fields, and actions arranged in a single accessible column without horizontal overflow.

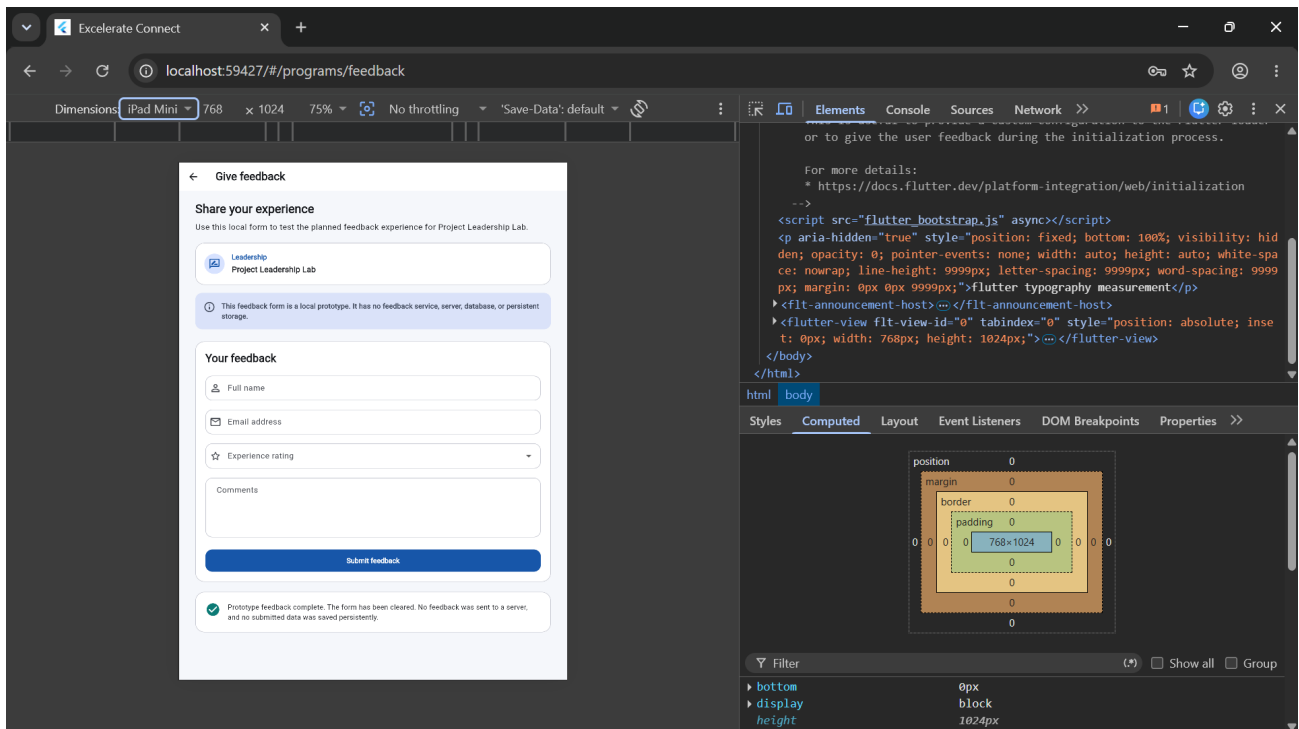
11. Responsive Large-Device Check



Caption: Feedback screen tested at approximately 853 x 1280.

The larger viewport preserves a centred content column, balanced spacing, readable form controls, and the complete success panel without stretching the interface excessively.

12. Responsive Tablet Check



Caption: Feedback screen tested at an iPad Mini viewport of approximately 768 x 1024.

The tablet layout retains the same content hierarchy and functional controls, confirming that the final responsive implementation adapts smoothly between mobile and desktop-sized screens.

Conclusion

Week 4 completes the four-week Excelerate Connect development cycle. The final prototype combines a responsive Flutter interface, bundled and validated JSON data, reusable program models, repository-driven loading, search and category filtering, selected-program navigation, validated registration and feedback forms, deliberate loading and submission states, accessibility refinements, and comprehensive testing evidence.

The application is suitable as a portfolio-ready local prototype and internship submission because its implemented features, test results, screenshots, and limitations are documented transparently. It does not present simulated behaviour as production functionality. The next technical phase would require a secure backend, verified live program data, real authentication, persistent submissions, and fully implemented learner and administrator workflows.

Final Completion Statement

The Week 4 final application was manually tested in Chrome, passed the reported automated validation checks, and is documented here with twelve final screenshots. No live backend, database, authentication service, API integration, deployment, or persistent data submission is claimed.

GitHub Repository: <https://github.com/franciskwarteng677/excelerate-connect>